Additional Problems

1. ZAMECH purchases subassemblies A and B from two suppliers located in Syracuse and New Orleans, respectively, for its assembly plants in Seattle and San Jose, which operate 360 days a year. The assembly plant in Seattle makes 50 products per day, whereas the plant in San Jose makes 75 products per day. Both types of products assembled in Seattle and San Jose require one unit of each of the subassemblies A and B. Trucks used have a capacity of 3,000 units and cost $4,000 per truckload shipped. The inventory holding cost per subassembly per day is $0.90, and the transit time associated with each route is 5 days.

a. Currently, direct shipments are made in full truckloads from the supplier locations to the assembly plants. Provide the total annual logistics (transportation, in-transit inventory, and on-site inventory) cost associated with the current system configuration.

Transportation Cost Syracuse to Seattle	$\frac{(4{,}000)(50)(360)}{3{,}000} = \$24{,}000/\text{year}$
Transportation Cost Syracuse to San Jose	$\frac{(4{,}000)(75)(360)}{3{,}000} = \$36{,}000/\text{year}$
Transportation Cost New Orleans to Seattle	$\frac{(4{,}000)(50)(360)}{3{,}000} = \$24{,}000/\text{year}$
Transportation Cost New Orleans to San Jose	$\frac{(4{,}000)(75)(360)}{3{,}000} = \$36{,}000/\text{year}$
Total Transportation Cost	$\$120{,}000/\text{year}$

In-transit Inv Cost Syracuse to Seattle	$(0.9)(360)(50)(360)\frac{5}{360} = \$81{,}000/\text{year}$
In-transit Inv Cost Syracuse to San Jose	$(0.9)(360)(75)(360)\frac{5}{360} = \$121{,}500/\text{year}$
In-transit Inv Cost New Orleans to Seattle	$(0.9)(360)(50)(360)\frac{5}{360} = \$81{,}000/\text{year}$
In-transit Inv Cost New Orleans to San Jose	$(0.9)(360)(75)(360)\frac{5}{360} = \$121{,}500/\text{year}$
Total In-transit Inv Cost	$\$405{,}000/\text{year}$

On-site Inv Cost Syracuse to Seattle	$\frac{(0.9)(360)(3{,}000)}{2} = \$486{,}000/\text{year}$
On-site Inv Cost Syracuse to San Jose	$\frac{(0.9)(360)(3{,}000)}{2} = \$486{,}000/\text{year}$
On-site Inv Cost New Orleans to Seattle	$\frac{(0.9)(360)(3{,}000)}{2} = \$486{,}000/\text{year}$
On-site Inv Cost New Orleans to San Jose	$\frac{(0.9)(360)(3{,}000)}{2} = \$486{,}000/\text{year}$
Total On-site Inv Cost	$\$1{,}944{,}000/\text{year}$

b. What would the total annual logistics (transportation, in-transit inventory, and on-site inventory) cost be, if optimal shipment quantities were to be used for making direct deliveries?

Optimal Shipment Size Syracuse to Seattle	$\sqrt{\frac{(2)(4{,}000)(50)(360)}{(0.9)(360)}} = 666.67 \text{ units}$
Optimal Shipment Size Syracuse to San Jose	$\sqrt{\frac{(2)(4{,}000)(75)(360)}{(0.9)(360)}} = 816.49\text{units}$
Optimal Shipment Size New Orleans to Seattle	$\sqrt{\frac{(2)(4{,}000)(50)(360)}{(0.9)(360)}} = 666.67 \text{ units}$
Optimal Shipment Size New Orleans to San Jose	$\sqrt{\frac{(2)(4{,}000)(75)(360)}{(0.9)(360)}} = 816.49\text{units}$

Transportation Cost Syracuse to Seattle	$\frac{(4{,}000)(50)(360)}{666.67} = \$107{,}999.46/\text{year}$
Transportation Cost Syracuse to San Jose	$\frac{(4{,}000)(75)(360)}{816.49} = \$132{,}273.51/\text{year}$
Transportation Cost New Orleans to Seattle	$\frac{(4{,}000)(50)(360)}{666.67} = \$107{,}999.46/\text{year}$
Transportation Cost New Orleans to San Jose	$\frac{(4{,}000)(75)(360)}{816.49} = \$132{,}273.51/\text{year}$
Total Transportation Cost	$\$480{,}545.94/\text{year}$

In-transit Inv Cost Syracuse to Seattle	$(0.40)(360)(100)(360)\frac{6}{360} = \$86{,}400/\text{year}$
In-transit Inv Cost Syracuse to San Jose	$(0.40)(360)(100)(360)\frac{6}{360} = \$86{,}400/\text{year}$
In-transit Inv Cost New Orleans to Seattle	$(0.40)(360)(100)(360)\frac{6}{360} = \$86{,}400/\text{year}$
In-transit Inv Cost New Orleans to San Jose	$(0.40)(360)(100)(360)\frac{6}{360} = \$86{,}400/\text{year}$
Total In-transit Inv Cost	$\$345{,}600.00/\text{year}$

On-site Inv Cost Syracuse to Seattle	$\frac{(0.9)(360)(666.67)}{2} = \$108{,}000.54/\text{year}$
On-site Inv Cost Syracuse to San Jose	$\frac{(0.9)(360)(816.49)}{2} = \$132{,}271.38/\text{year}$
On-site Inv Cost New Orleans to Seattle	$\frac{(0.9)(360)(666.67)}{2} = \$108{,}000.54/\text{year}$
On-site Inv Cost New Orleans to San Jose	$\frac{(0.9)(360)(816.49)}{2} = \$132{,}271.38/\text{year}$
Total On-Site Inv Cost	$\$480{,}543.84/\text{year}$

c. Consider the following milk-run pick-up and delivery scheme: Suppose that a truck picks up subassemblies first from Syracuse, then from New Orleans and drops them off first at San Jose and then at Seattle. What would the total annual logistics (transportation, in-transit inventory, and on-site inventory) cost be, if optimal shipment quantity for this milk-run delivery scheme were to be used? In calculating the in-transit inventory holding cost associated with this system configuration, you may assume that it takes exactly 3 days to drive from Syracuse to New Orleans, exactly 3 days from New Orleans to San Jose and exactly 3 days from San Jose to Seattle.

Optimal Shipment Size from Syracuse to New Orleans to San Jose to Seattle	$\sqrt{\frac{(2)(4{,}000)(50+50+75+75)(360)}{(0.9)(360)}} = 1{,}490.71 \text{ units}$

Transportation Cost from Syracuse to New Orleans to San Jose to Seattle	$\frac{(4{,}000)(50+50+75+75)(360)}{1{,}490.71} = \$241{,}495.66/\text{year}$

In-transit Inv Cost Syracuse to New Orleans	$(0.9)(360)(50+75)(360)\frac{3}{360} = \$121{,}500/\text{year}$
In-transit Inv Cost New Orleans to San Jose	$(0.9)(360)(50+75+50+75)(360)\frac{3}{360} = \$243{,}000/\text{year}$
In-transit Inv Cost San Jose to Seattle	$(0.9)(360)(50+50)(360)\frac{3}{360} = \$97{,}200/\text{year}$
Total In-Transit Inv Cost	$\$461{,}700/\text{year}$

On-site Inv Cost San Jose and Seattle	$\frac{(0.9)(360)(1{,}490.71)}{2} = \$241{,}495.02/\text{year}$

2. IKEA operates 8 stores on the West Coast that are replenished every six weeks from the central distribution center located in New Jersey. The weekly demand for Poang chairs at these store locations are identically and independently distributed and can be assumed to follow a normal distribution with a mean of 14 chairs per week and a variance of 16 chairs^2. The chairs weigh 10 lbs. and cost $79 each. The annual inventory carrying rate for the West Coast stores is 52 percent.

IKEA outsources its distribution activities to TNT Logistics. The delivery lead time from New Jersey to any of the stores on the West Coast is approximately normally distributed with a mean of 3 weeks and a standard deviation of 1 week. TNT charges IKEA a transportation rate of $\$250 + 4x$, where x is the weight of the shipment in pounds. Assume that there are 52 weeks in a year.

IKEA would like to maintain enough safety stock in each of its retail stores such that the probability of not stocking out during delivery lead time is 96 percent.

a. What is the total annual logistics cost associated with the current system configuration?

Shipment quantity for one store	$q_i = (6)(14) = 84$ unis
Average demand during lead time	$\mu_L = (3)(14) = 42$ units
Std Dev of demand during lead time	$\sigma_L = \sqrt{(3)(4^2) + (14^2)(1^2)} = 15.62$ units
Type I service level	$\alpha = F(r) = 0.96 \Rightarrow z = 1.7507$
Safet stock quantity	$s_i = (15.62)(1.7507) = 27.35$ units
Transportation cost	$[250 + (4)(84)(10)]\frac{(52)(14)}{84} = \$31{,}286.67/\text{year}$
On-site inventory holding cost	$(0.01)(52)(79)\left(27.35 + \frac{84}{2}\right) = \$2{,}848.77/\text{year}$
In-transit inventory holding cost	$(0.01)(52)(79)(52)(14)\left(\frac{3}{52}\right) = \$1{,}725.36/\text{year}$
Total cost for one store	$\$35{,}860.79/\text{year}$
Total cost for the entire network	$(8)(35{,}860.79) = \$286{,}886.3/\text{year}$

b. TNT Logistics offers a guaranteed constant delivery lead time of one week at a rate of $\$300 + 5x$, where x is the weight of the shipment in pounds. Should IKEA accept this offer?

Shipment quantity for one store	$q_i = (6)(14) = 84$ unis
Average demand during lead time	$\mu_L = (1)(14) = 14$ units
Std Dev of demand during lead time	$\sigma_L = \sqrt{(1)(4^2)} = 4$ units
Type I service level	$\alpha = F(r) = 0.96 \Rightarrow z = 1.7507$
Safet stock quantity	$s_i = (1.7507)(4) = 7.00$ units
Transportation cost	$[300 + (4)(84)(10)]\frac{(52)(14)}{84} = \$31{,}720/\text{year}$
On-site inventory holding cost	$(0.01)(52)(79)\left(7.00 + \frac{84}{2}\right) = \$2{,}013.03/\text{year}$
In-transit inventory holding cost	$(0.01)(52)(79)(52)(14)\left(\frac{1}{52}\right) = \$575.12/\text{year}$
Total cost for one store	$\$34{,}308.15/\text{year}$
Total cost for the entire network	$(8)(34{,}308.15) = \$274{,}465.2/\text{year}$

IKEA should accept this offer.

3. Emu Inc., is considering two truck services to make deliveries from its manufacturing plant to its central warehouse. Orange Transportation is cheaper but is also slower and less reliable than Blue Transportation. The following information is available:

Average daily demand (units)	80

Std Dev of daily demand (units)	12
Fixed ordering cost ($/order)	150
Unit product price ($/unit)	50
Shipment size (units)	As per EOQ
In-transit inventory carrying rate	30% per year
On-site inventory carrying rate	20% per year
In-stock probability during lead time	90
Unit stock-out cost	Unknown
Sellin days	360 days/year

	Orange Transportation	Blue Transportation
Average Transit Time (days)	7	3
Std Dev of Transit Time (days)	1.8	1.2
Rate ($/unit)	11.45	11.55

A lot-size-reorder-point inventory policy is being used for planning at the warehouse. From the point of view of total annual inventory- and transportation-related costs, which truck service should be selected?

Orange Transportation:

Shipment quantity	$q_{OT} = \sqrt{\frac{(2)(150)(80)(360)}{(0.20)(50)}} = 929.52$ unis
Average demand during lead time	$\mu_L^{OT} = (7)(80) = 560$ units
Std Dev of demand during lead time	$\sigma_L^{OT} = \sqrt{(7)(12^2) + (80^2)(1.8^2)} = 147.46$ units
Type I service level	$\alpha = F(r) = 0.90 \Rightarrow z = 1.2816$
Safet stock quantity	$s^{OT} = (147.46)(1.2816) = 188.75$ units
Ordering cost	$(150)\left(\frac{(80)(360)}{929.52}\right) = \$4{,}547.56/\text{year}$
Transportation cost	$(11.45)(80)(360) = \$329{,}760/\text{year}$
On-site inventory holding cost	$(0.20)(50)\left(188.75 + \frac{929.53}{2}\right) = \$6{,}535.10/\text{year}$
In-transit inventory holding cost	$(0.30)(50)(80)(360)\left(\frac{7}{360}\right) = \$8{,}400/\text{year}$
Total cost using Orange	$\$349{,}242.65/\text{year}$

Orange Transportation:

Shipment quantity	$q_{BT} = \sqrt{\frac{(2)(150)(80)(360)}{(0.20)(50)}} = 929.52$ unis
Average demand during lead time	$\mu_L^{BT} = (3)(80) = 240$ units
Std Dev of demand during lead time	$\sigma_L^{BT} = \sqrt{(3)(12^2) + (80^2)(1.2^2)} = 98.22$ units
Type I service level	$\alpha = F(r) = 0.90 \Rightarrow z = 1.2816$
Safet stock quantity	$s^{BT} = (98.22)(1.2816) = 125.73$ units
Ordering cost	$(150)\left(\frac{(80)(360)}{929.52}\right) = \$4{,}547.56/\text{year}$
Transportation cost	$(11.55)(80)(360) = \$332{,}640/\text{year}$
On-site inventory holding cost	$(0.20)(50)\left(125.73 + \frac{929.52}{2}\right) = \$5{,}904.9/\text{year}$
In-transit inventory holding cost	$(0.30)(50)(80)(360)\left(\frac{3}{360}\right) = \$3{,}600/\text{year}$
Total cost using Blue	$\$346{,}692.46/\text{year}$

Decision: Blue Transportation should be chosen.

4. RuenTex purchases components from two suppliers located in Austin and Carthage. The supplier in Austin provides component A and the one in Carthage provides component B. These components are used to make final assemblies of two types, Assembly I and Assembly II. Assembly I is made in Helena, and Assembly II in Duluth. Both assemblies require one unit of each of the components A and B. Each assembly plant makes 50 assemblies per day.

Currently, the components are shipped directly in full truckloads from the suppliers to the assembly plants. The trucks used have a capacity of 500 units of either component, and a cost of $2,500 per truckload. The inventory holding cost per component is $0.50 per day, and the transit time per route is 4 days. Assume that there are 360 days in a year.

a. What is the total annual logistics cost for this system configuration?

Transportation Cost Austin to Helena	$\frac{(2{,}500)(50)(360)}{500} = \$90{,}000/\text{year}$
Transportation Cost Austin to Duluth	$\frac{(2{,}500)(50)(360)}{500} = \$90{,}000/\text{year}$
Transportation Cost Carthage to Helena	$\frac{(2{,}500)(50)(360)}{500} = \$90{,}000/\text{year}$
Transportation Cost Carthage to Duluth	$\frac{(2{,}500)(50)(360)}{500} = \$90{,}000/\text{year}$
Total Transportation Cost	$\$360{,}000/\text{year}$

In-transit Inv Cost Austin to Helena	$(0.5)(360)(50)(360)\frac{4}{360} = \$36{,}000/\text{year}$
In-transit Inv Cost Austin to Duluth	$(0.5)(360)(50)(360)\frac{4}{360} = \$36{,}000/\text{year}$
In-transit Inv Cost Carthage to Helena	$(0.5)(360)(50)(360)\frac{4}{360} = \$36{,}000/\text{year}$
In-transit Inv Cost Carthage to Duluth	$(0.5)(360)(50)(360)\frac{4}{360} = \$36{,}000/\text{year}$
Total In-transit Inv Cost	$\$144{,}000/\text{year}$

On-site Inv Cost Austin to Helena	$\frac{(0.5)(360)(500)}{2} = \$45{,}000\text{year}$
On-site Inv Cost Austin to Duluth	$\frac{(0.5)(360)(500)}{2} = \$45{,}000\text{year}$
On-site Inv Cost Carthage to Helena	$\frac{(0.5)(360)(500)}{2} = \$45{,}000\text{year}$
On-site Inv Cost Carthage to Duluth	$\frac{(0.5)(360)(500)}{2} = \$45{,}000\text{year}$
Total On-site Inv Cost	$180{,}000/\text{year}$

b. RuenTex has done a market research and found out that larger trucks with a capacity of 1,000 units can be obtained at a cost of $3,500 per route and a transit time of 4 days per route. If RuenTex decides to use these trucks, should all shipments be full truckloads or less than full truckloads? What would the total annual logistics cost be, if RuenTex were to use these larger trucks with the appropriate level of load?

Optimal Shipment Size Austin to Helena	$\sqrt{\frac{(2)(3{,}500)(50)(360)}{(0.5)(360)}} = 836.66 \text{ units}$
Optimal Shipment Size Austin to Duluth	$\sqrt{\frac{(2)(3{,}500)(50)(360)}{(0.5)(360)}} = 836.66 \text{ units}$
Optimal Shipment Size Carthage to Helena	$\sqrt{\frac{(2)(3{,}500)(50)(360)}{(0.5)(360)}} = 836.66 \text{ units}$
Optimal Shipment Size Carthage to Duluth	$\sqrt{\frac{(2)(3{,}500)(50)(360)}{(0.5)(360)}} = 836.66 \text{ units}$

Note that the optimal shipment size is less than the truck capacity, i.e., $836.66 \text{ units} < 1{,}000 \text{ units}$. Hence, it must be the optimal policy to use the optimal shipment sizes as per EOQ. Hence, we have the following:

Transportation Cost Austin to Helena	$\frac{(3{,}500)(50)(360)}{500} = \$75{,}299.41/\text{year}$

Transportation Cost Austin to Duluth	$\frac{(3{,}500)(50)(360)}{500} = \$75{,}299.41/\text{year}$
Transportation Cost Carthage to Helena	$\frac{(3{,}500)(50)(360)}{500} = \$75{,}299.41/\text{year}$
Transportation Cost Carthage to Duluth	$\frac{(3{,}500)(50)(360)}{500} = \$75{,}299.41/\text{yearr}$
Total Transportation Cost	$\$301{,}197.6/\text{year}$

In-transit Inv Cost Austin to Helena	$(0.5)(360)(50)(360)\frac{4}{360} = \$36{,}000/\text{year}$
In-transit Inv Cost Austin to Duluth	$(0.5)(360)(50)(360)\frac{4}{360} = \$36{,}000/\text{year}$
In-transit Inv Cost Carthage to Helena	$(0.5)(360)(50)(360)\frac{4}{360} = \$36{,}000/\text{year}$
In-transit Inv Cost Carthage to Duluth	$(0.5)(360)(50)(360)\frac{4}{360} = \$36{,}000/\text{year}$
Total In-transit Inv Cost	$\$144{,}000/\text{year}$

On-site Inv Cost Austin to Helena	$\frac{(0.5)(360)(836.66)}{2} = \$75{,}299./\text{year}$
On-site Inv Cost Austin to Duluth	$\frac{(0.5)(360)(836.66)}{2} = \$75{,}299./\text{year}$
On-site Inv Cost Carthage to Helena	$\frac{(0.5)(360)(836.66)}{2} = \$75{,}299./4\text{year}$
On-site Inv Cost Carthage to Duluth	$\frac{(0.5)(360)(836.66)}{2} = \$75{,}299.4/\text{year}$
Total On-site Inv Cost	$301{,}197.6/\text{year}$

c. RuenTex has the option of using a warehouse in Kansas City to store the components incurring an inventory holding cost of $0.40 per day per component. In this case, trucks (with 1,000-unit capacity) would go from Austin to Kansas City to drop off full truckloads of component A with a transit time of 2 days. Similarly, trucks (with 1,000-unit capacity) would go from Carthage to Kansas City to drop off full truckloads of component B with a transit time of 2 days. The trucks from Austin and Carthage would arrive at the warehouse in Kansas City on the same day. Every 5 days a truck would leave from the warehouse in Kansas City and drop off equal-sized loads in Helena and Duluth with a total transit time of 2 days. Assume that the cost for each of the routes (from Austin to Kansas City, from Carthage to Kansas City and from Kansas City to Helena and Duluth) is $2,000 per route. What would the total annual logistics cost associated with this system be? Note: In calculating the in-transit inventory holding cost from Kansas City to Helena and Duluth you may assume that the entire load on the truck spends two days in transportation.

Transportation Cost Austin to Kansas City	$\frac{(2{,}000)(50+50)(360)}{1{,}000} = \$72{,}000/\text{year}$

Transportation Cost Carthage to Kansas City	$\frac{(2{,}000)(50+50)(360)}{1{,}000} = \$72{,}000/\text{year}$
Transportation Cost Kansas City to Helena to Duluth	$\frac{(2{,}000)(50+50+50+50)(360)}{1{,}000} = \$144{,}000/\text{year}$
Total Transportation Cost	$\$288{,}000/\text{year}$

In-transit Inv Cost Austin to Kansas City	$(0.5)(360)(50+50)(360)\frac{2}{360} = \$36{,}000/\text{year}$
In-transit Inv Cost Carthage to Kansas City	$(0.5)(360)(50+50)(360)\frac{2}{360} = \$36{,}000/\text{year}$
In-transit Inv Cost Kansas City to Helena and Duluth	$(0.5)(360)(50+50+50+50)(360)\frac{2}{360} = \$72{,}000/\text{year}$
Total In-transit Inv Cost	$\$144{,}000/\text{year}$

On-site Inv Cost Helena	$\frac{(0.5)(360)(500)}{2} = \$45{,}000/\text{year}$
On-site Inv Cost Duluth	$\frac{(0.5)(360)(500)}{2} = \$45{,}000/\text{year}$
On-site Inv Cost Kansas City	$(0.4)(360)(1{,}000)\left(\frac{5}{360}\right)\left(\frac{360}{10}\right) = \$72{,}000/\text{year}$
Total On-site Inv Cost	$\$162{,}000/\text{year}$

In the on-site inventory holding cost above note the following. Full truckload deliveries of component A and B bring 1000 units of component A and 1000 units of component B. When a truck with a capacity of 1,000 units leave to make deliveries to Helena and Duluth, 1000 units (500 units of component A and 500 units of component B) are left behind at the warehouse in Kansas City until the next delivery. Hence, the inventory profile at Kansas City mimics a pulse function, that is at 0 unit for 10 days, and is at 1,000 units for the next 10 days through the year. The area under this curve needs to be calculate to evaluate the annual on-site inventory holding cost at Kansas City.

d. If Kansas City were to be converted into a pure cross-docking facility, how should the shipments to and from Kansas City be adjusted? What would the total annual logistics cost for this system configuration be?

Note that there are multiple ways to operate Kansas City as a cross-docking facility. One way is to make direct shipments of pure loads (containing component A from Austin or B) into Kansas City and to make direct shipments of mixed loads (containing components A and B) from Kansas City.

Transportation Cost Austin to Kansas City	$\frac{(2{,}000)(50+50)(360)}{1{,}000} = \$72{,}000/\text{year}$
Transportation Cost Carthage to Kansas City	$\frac{(2{,}000)(50+50)(360)}{1{,}000} = \$72{,}000/\text{year}$

Transportation Cost Kansas City to Helena	$\frac{(2{,}000)(50+50)(360)}{1{,}000} = \$72{,}000/\text{year}$
Transportation Cost Kansas City to Helena	$\frac{(2{,}000)(50+50)(360)}{1{,}000} = \$72{,}000/\text{year}$
Total Transportation Cost	$\$288{,}000/\text{year}$

In-transit Inv Cost Austin to Kansas City	$(0.5)(360)(50+50)(360)\frac{2}{360} = \$36{,}000/\text{year}$
In-transit Inv Cost Carthage to Kansas City	$(0.5)(360)(50+50)(360)\frac{2}{360} = \$36{,}000/\text{year}$
In-transit Inv Cost Kansas City to Helena	$(0.5)(360)(50+50)(360)\frac{2}{360} = \$36{,}000/\text{year}$
In-transit Inv Cost Kansas City to Duluth	$(0.5)(360)(50+50)(360)\frac{2}{360} = \$36{,}000/\text{year}$
Total In-transit Inv Cost	$\$144{,}000/\text{year}$

On-site Inv Cost Kansas City	$\$0/\text{year}$
On-site Inv Cost Duluth	$\frac{(0.5)(360)(1{,}000)}{2} = \$90{,}000/\text{year}$
On-site Inv Cost Helena	$\frac{(0.5)(360)(1{,}000)}{2} = \$90{,}000/\text{year}$
Total On-site Inv Cost	$\$180{,}000/\text{year}$

Note that alternatively, a milk-run pick-up scheme can be used to bring components to Kansas City and a milk-run pick-up scheme can be used to deliver the components to assembly facilities. The answer would change depending on how one chooses to coordinate the shipments and operate the system as a pure cross-docking facility.